The empirical recurrence for OEIS A240366: recovering a conjecture that is not written down

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Abstract

OEIS A240366 records "Empirical recurrence of order 80", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 124$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 80, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 80 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A240366 is "Number of 3Xn 0..3 arrays with no element equal to one plus the sum of elements to its left or two plus the sum of the elements above it or two plus the sum of the elements diagonally to its northwest, modulo 4.". It has offset 1 and begins

$$7, 43, 257, 1693, 10997, 74255, 492758, 3349106, \ \ldots$$

The entry states:

> Empirical recurrence of order 80 (see link above)

As of the "Last modified" line on the live entry (Jun 02 2025 09:51:18, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 124$ of the 256, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 124.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 248$ exact terms of the model returns order 80 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{80} c_i \, a(n-i),$$

c_{1}	8	c_{21}	3258826705	c_{41}	311024576895	c_{61}	−11555386526707
c_{2}	52	c_{22}	−2973387241	c_{42}	−163262879798	c_{62}	4206652476337
c_{3}	−527	c_{23}	−8194146262	c_{43}	1842528177460	c_{63}	9447251803171
c_{4}	−552	c_{24}	19523134472	c_{44}	−274885928633	c_{64}	−6821481913164
c_{5}	12560	c_{25}	5607713436	c_{45}	−3005236487017	c_{65}	−4662833495886
c_{6}	−7920	c_{26}	−40873689296	c_{46}	−138289726724	c_{66}	5390611869473
c_{7}	−147919	c_{27}	23843244062	c_{47}	1668883038999	c_{67}	480291937762
c_{8}	232662	c_{28}	22390420517	c_{48}	2456017017769	c_{68}	−2547644091294
c_{9}	890118	c_{29}	−88011218639	c_{49}	1483740037317	c_{69}	829767570866
c_{10}	−2229550	c_{30}	70648129219	c_{50}	−8036508242068	c_{70}	634157227952
c_{11}	−1696482	c_{31}	178332325951	c_{51}	−2127901359355	c_{71}	−245111998652
c_{12}	9553892	c_{32}	−158366143019	c_{52}	12644141808563	c_{72}	−53739829476
c_{13}	−11958797	c_{33}	−289176114443	c_{53}	−2266382370236	c_{73}	21362320188
c_{14}	−1853450	c_{34}	148011508255	c_{54}	−7005060866344	c_{74}	−28120408
c_{15}	89426662	c_{35}	226924384926	c_{55}	5369918347133	c_{75}	9770945520
c_{16}	−179534657	c_{36}	−101382787278	c_{56}	−3671908870004	c_{76}	2147048528
c_{17}	−191684800	c_{37}	427046973892	c_{57}	−5231562047225	c_{77}	−1274905952
c_{18}	849066746	c_{38}	−60250661524	c_{58}	5136108201116	c_{78}	−8738496
c_{19}	−370826958	c_{39}	−1145037080934	c_{59}	8452368104418	c_{79}	5495040
c_{20}	−1330916663	c_{40}	343907223624	c_{60}	−2121586831231	c_{80}	8508416

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 80, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $80 + 4$ terms removed returns order 80 as well, so $\deg q_{\min} = 80 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $80+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 80 that this sequence satisfies — and that family turns out to have exactly one member.

References

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